Harsh Vaishnav

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LinkedIn | GitHub

Education

Indian Institute of Information Technology (IIIT), Nagpur **Aug 2023 – May 2027**
Bachelor of Technology in Computer Science and Engineering

Technical Skills

Languages	C++, Python, JavaScript, Java, C, SQL
Core CS	Object-Oriented Programming, Operating Systems, DBMS, Computer Networks
Tools & Frameworks	Git, GitHub, React, Node.js, Express.js, FastAPI, Socket.IO, Streamlit, REST APIs, JWT Auth
AI/ML & LLM APIs	LangChain, LangGraph, LangSmith, FAISS, RAG, Mistral AI, Google Gemini, Groq, Tavily
Databases	MongoDB, MySQL

Experience

Global Finalist – HackIndia Hackathon **Jan 2025**
India's Biggest AI & Web3 Hackathon · 29000+ Participants

- Advanced to the Global Finals among 29000+ participants by building **summary.ai** — an **Agentic RAG Document Intelligence Platform** (FastAPI + React) with JWT-secured, multi-tenant architecture.
- Engineered a multi-agent query system with distinct **Search, Explore, and Think** pipelines: **FAISS + MMR** vector retrieval for semantic search, a **LangChain agent tool** for social data exploration, and a **Tavily + BeautifulSoup** web-scraping pipeline for live augmentation, synthesized via **Mistral AI and Google Gemini**.
- Presented the live, end-to-end system to **industry judges from AI & Web3 domains**; recognized among the top teams globally out of 29000+ participants.

Projects

Harsh Bank of India

- Built a modular banking backend using **Node.js, Express.js, and MongoDB** with a **double-entry ledger architecture**, deriving account balance via **MongoDB aggregation** ($match, $group, $sum) and enforcing immutability via **Mongoose pre-hooks** blocking direct mutation.
- Implemented atomic fund transfers using **MongoDB ClientSession transactions** with **idempotency-key handling** to prevent duplicate charges, alongside balance/ownership validation and transaction status tracking.
- Architected a **LangGraph multi-agent AI service** (FastAPI) with a **Teller-Underwriter state-machine**, isolating conversational access from a backend-only loan-disbursement agent to mitigate prompt-injection risk, with **LangSmith-traced** observability and thread-scoped context persistence via **MemorySaver**.

VideoMind – AI Video/Meeting Assistant

- Built an audio ingestion pipeline (**yt-dlp, pydub, ffmpeg**) that downloads YouTube audio or converts local files to WAV, auto-chunking long recordings, then routed transcription across two paths – **local Whisper** for English and the **Sarvam AI STT-translate API** for Hinglish.
- Designed a RAG pipeline using **LangChain LCEL**, chunking transcripts into a persisted **ChromaDB vector store** with **HuggingFace MiniLM embeddings**, and grounding a **Mistral AI-backed retrieval chain** to answer meeting questions strictly from transcript context.
- Built structured-extraction chains (action items with owner/deadline, key decisions, open questions) and a **Streamlit UI** with live per-stage pipeline status tracking (audio → transcript → summary → RAG indexing).

Real-Time Collaborative Code Editor

- Built a Google Docs-style collaborative code editor (**React.js, Monaco Editor, Node.js, Express.js, Socket.IO**) supporting simultaneous multi-user editing with live cursors and presence.
- Implemented conflict-free real-time sync using **Yjs (CRDTs)** and the **Awareness Protocol**, merging concurrent edits and broadcasting cursor/selection state over WebSockets.

Achievements

- **3rd Place, Rebranding Wars Hackathon** – Rebranded Zepto's visual identity (logo, palette, typography) in a 6-hour sprint.
- **LeetCode – Max Rating: 1851**. Solved **290+ problems** across data structures, dynamic programming, and graphs.
- **CodeChef – Max Rating: 1611 (3 Star)**. Maintained consistent performance across **40+ rated contests**.
- **Codeforces – Pupil (1240)**. Participated in **35+ rated rounds**, strengthening problem-solving speed under contest conditions.
- Solved **850+ DSA problems** across LeetCode, Codeforces, CodeChef, CSES, and NeetCode.

Leadership & Activities

- **Event Head, Flash Code Coding Contest (Dot-Slash Coding Club)**: Organized a coding contest with **500+ registrations**, selecting **40 finalists**; coordinated a **team of 10 volunteers** across problem setting, logistics, and evaluation.
- **Mentor, Winter of Code (Dot-Slash Coding Club)**: Guided **5–6 students** in Data Structures & Algorithms.
- **Core Member, Dot-Slash Coding Club**: Selected among the **top 10 members**; contributed to mentoring students.